Quantum Ledger Technology (QLT): Review of A Blockchain-Agnostic Framework for Securing DeFi and Web 3.0 Infrastructure in the Post-Quantum Era

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Structural Layout

- Future directions: federation, standards alignment, open-source reference implementations.

1. Introduction

Charlie Bell, Microsoft Cybersecurity head, once said, "Cybersecurity is the mother of all problems," that unless the security problem is solved, innovation in other areas is impossible because every new technology can be weaponized against us.[1] This essentially means that our computing systems, without exception, remain vulnerable to adversarial attacks. This also means that our computing systems will always have an attack surface that can only be patched, not eliminated, at design time. The latest example is Anthropic's launch of Opus 4.6 on February 5, 2026, when its Frontier Red Team reported that Opus 4.6 autonomously identified and validated more than **500 high-severity zero-day vulnerabilities** in widely used open-source libraries that are deployed by almost all software developers. The panic caused the market crash to erase a massive $830 billion from the market capitalization of the top software companies.[2] AI has created an environment where traditional security is no longer sufficient for business continuity.[3] Quantum Computing (QC) represents another paradigm shift with profound implications for digital security and economic stability. Recent breakthroughs, such as Google's claim of a quantum computer 241 million times faster than its 2019 predecessor,[4] and China's Zuchongzhi-3 achieving trillions-of-times-faster speeds, [5] signal that the era of encryption-breaking QC is imminent.[6] Within the next 5–10 years, current cryptographic standards based on RSA and Elliptic Curve Cryptography (ECC) are expected to be rendered obsolete by Shor's and Grover's algorithms.[7] This "Q-Day" scenario poses an existential threat to the $3+ trillion DeFi economy, as well as to Web 3.0 infrastructures that rely entirely on user-facing cryptography.[8],[9] Experts project 2030 as the year when encryption-breaking QC becomes available (**Fig. 1**).[10] This creates a QC threat paradox (**Fig. 2**).

Fig.1. 2030-Projected year crypto-breaking QC becomes available (fn1).

The global response has largely centered on Post-Quantum Cryptography (PQC). Since 2016, the National Institute of Standards and Technology (NIST) has attempted to standardize quantum-resistant algorithms.[11] Yet, despite years of effort, most candidate PQCs have failed under cryptanalysis, and those that

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[1]Alex Halverson. Microsoft's cybersecurity chief weighs in on 'mother of all problems.' The Business Journal, October 6, 2022. Available at https://www.bizjournals.com/seattle/news/2022/10/06/microsoft-charlie-bell-cybersecurity-geekwire.html

[2]Novy Baf. $830 Billion Vanished in One Week - Because This AI Found 500 Security Flaws Nobody Else Could. The Novo Tech, February 8, 2026. Available: https://www.thenovtech.com/p/830-billion-vanished-in-one-week

[3] Awan, M., Alam, A., Khan, R.A. *et al.* A generative AI-driven cybersecurity framework for small and medium enterprises software. development: an ANN-ISM approach. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-37614-8

[4] Carnell, Mike. "Buckle Up." *Quality Progress* 58.3 (2025).

[5] Gao, Dongxin, et al. "Establishing a new benchmark in quantum computational advantage with 105-qubit Zuchongzhi 3.0 processor." *Physical Review Letters* 134.9 (2025): 090601.

[6] Salampasis, Dimitrios. "Preparing for the Quantum Computing Revolution." *Research-Technology Management* 68.5 (2025): 69-72.

[7] Zimmermann, Daniel. "Adoption of Post-Quantum Cryptography in Organizations: Challenges and Drivers." *Future* 36.1: 256-280.

[8] Qader, Timur. *Systematic Security: Building Quantum Security: Preparing for Q-Day and Beyond*. CRC Press, 2026. https://doi.org/10.1201/9781003685746

[9] Schiffer, Benjamin F. "Quantum computers as an amplifier for existential risk." *arXiv preprint arXiv:2205.02761* (2022).

[10]GQI. Cloud Security Alliance Creates a Countdown to Y2Q Clock. Quantum Computing Report (undated). Available at https://quantumcomputingreport.com/cloud-security-alliance-creates-a-countdown-to-y2q-clock/

[11] Dua, Aarushi, Anubhav Gupta, and Bhawana Pujasani Hettiarachchi. "Quantum resistant cryptography: Survey of NIST third round candidates and real world implementation." *AIP Conference Proceedings*. Vol. 3297. No. 1. AIP Publishing LLC, 2025.

remain are computationally expensive, latency-prone, and unsuitable for resource-constrained environments such as IoT devices or high-throughput blockchain networks. PQC's fragility underscores the need for an alternative approach, one that is architectural rather than algorithmic.
Quantum Ledger Technology (QLT) emerges as such an approach. Conceived as a blockchain-agnostic framework, QLT integrates Zero Vulnerability Computing (ZVC)[12] and Solid-State Software on a Chip (3SoC) to eliminate third-party permissions and deliver a zero-attack-surface environment. [13] Because ZVC eliminates all third-party permissions, it does not require implementing or monitoring policies to regulate non-native or adversary code to achieve a zero-trust architecture.[14] Absolute Zero Trust (AZT) is achieved automatically and by default. Unlike PQC, which seeks to harden exposed cryptographic primitives, QLT separates blockchain infrastructure from the mainstream Internet, creating a closed, quantum-resilient intranet for exchanges, wallets, and decentralized applications.[15] Evolving from the first principles of software engineering,[16] this architectural shift neutralizes both legacy hacks and quantum adversaries. Conceptually, because the ZVC architecture does not deploy user-facing classical or quantum cryptography, its access authentication is inherently ultra-low-latency.[17]

Access authentication, which remains the epicenter of any cybersecurity strategy, constitutes a critical enhancement to QLT through the introduction of Quantum-Resilient User-Evasive Cryptographic Authentication (QRUECA)**.**[18] Traditional authentication relies on user-facing cryptography, passwords, private keys, and biometrics, all of which expose attack surfaces that quantum adversaries can exploit. QRUECA replaces these with a multi-gate, device-to-device handshake sealed within hardware, eliminating user-visible credentials and rendering authentication inherently quantum-resilient. By embedding QRUECA into QLT, the framework not only secures transactions but also redefines trust in Web 3.0 ecosystems.
This paper presents QLT with QRUECA as a unified, deployable solution for securing DeFi, Web 3.0, and emerging infrastructures such as 6G networks. It argues that architectural resilience, rather than cryptographic patchwork, is the only viable path to sustainable security in the post-quantum era. The remainder of the paper is structured as follows: Section 2 reviews blockchain vulnerabilities and PQC limitations; Section 3 outlines the quantum threat landscape; Section 4 details the QLT framework; Section 5 introduces QRUECA as its authentication protocol; Section 6 discusses operational benefits; Section 7 explores applications across DeFi, Web 3.0, and 6G; Section 8 provides comparative analysis; and Section 9 concludes with future directions.

2. Background and Related Work

2.1 Blockchain Vulnerabilities

Since its inception with Bitcoin in 2008, blockchain has been heralded as a secure, decentralized infrastructure. Yet the ecosystem has been plagued by persistent vulnerabilities. Centralized exchanges have incurred billions in losses due to custodial breaches, whereas decentralized exchanges have been compromised by cross-chain bridge exploits. Smart contracts, often written in Turing-complete languages

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12 Raheman, Fazal, et al. "Will zero vulnerability computing (ZVC) ever be possible? testing the hypothesis." *Future Internet* 14.8 (2022): 238.

13 Raheman, Fazal. "The future of cybersecurity in the age of quantum computers." *Future Internet* 14.11 (2022): 335.

14 F..Raheman, From Standard Policy-Based Zero Trust to Absolute Zero Trust (AZT): A Quantum Leap to Q-Day Security. *J Comp and Communications*, 12, (2024) 252-282.

15 Raheman, Fazal. "Futureproofing Blockchain & Cryptocurrencies against Growing Vulnerabilities & Q-Day Threat with Quantum-Safe Ledger Technology (QLT)." *Journal of Cmputer and Communications* 12.7 (2024): 59-77.

16 F Raheman, et al. A First Principal Review of a Novel De Facto Preventive Cybersecurity Strategy. (under peer review). Available at www.bc5.eu/prosec.pdf

17 F. Raheman, "Formulating & supporting a hypothesis to address a Catch-22 situation in 6G communication networks," *J. of Info. Security*, vol15:3,340–354, 2024.

18 Ali Raheman, Tejas Bhagat, F Raheman. Quantum-Resilient User-Evasive Cryptographic Authentication (QRUECA) for Web 3.0 Security in the Post-Quantum Era. Proceedings of QC-Horizon Conference, August 26-27, 2025, Pilsen, Czech Republic (2025)

such as Solidity, remain prone to coding errors and logic flaws, enabling adversaries to manipulate contract execution. These vulnerabilities introduce systemic volatility into cryptocurrency markets, eroding trust and slowing adoption. The reliance on user-facing cryptography, private keys, passwords, and signatures further exposes blockchain systems to phishing, credential theft, and replay attacks.

Table 1. Summary of cryptocurrency hacks since the launch of Bitcoin

Scope	Estimated Loss
Hacks & scams since ~2011	≈ **$22.7 bn**[19]
Scams & fraud in 2025 alone	≈ **$17 bn**[20]
Hacks & scams annual totals (recent years)	~$2–3 bn per year range before 2025

While detailed totals vary, credible reports place *cumulative cryptocurrency hacking, scam, and fraud losses over the history of Bitcoin at* ***tens of billions*** *of dollars*, with 2025 being a particularly extreme year.

2.2 Post-Quantum Cryptography (PQC) Limitations

The looming quantum threat compounds these vulnerabilities. Shor's algorithm threatens RSA and ECC, while Grover's algorithm halves the effective strength of symmetric keys. In response, NIST initiated a PQC standardization process in 2016.[21] However, most candidate algorithms have failed under cryptanalysis, and those that remain are computationally expensive, requiring large key sizes and introducing latency. PQC's resource intensity makes it unsuitable for the majority of IoT devices (estimated at 70–85%), which are resource-constrained endpoints relying on low-power microcontrollers with limited memory & compute capacity (**Fig. 3**). As a result, they are unable to support conventional cryptographic or trust management mechanisms. As confirmed by peer-reviewed studies, Class 0 & Class 1 IoT devices lack the memory & compute capacity to support conventional cryptographic protocols or trust management systems.[22],[23] This reinforces the need for credential-agnostic authentication & a QC-resilient safety layer that controls the lightning-fast QC's decryption power. Moreover, PQC addresses algorithmic resilience but does not mitigate architectural vulnerabilities that create the attack surface by including third-party permissions (TPPs) and user-facing cryptographic exposure. This gap underscores the need for a paradigm shift beyond PQC.

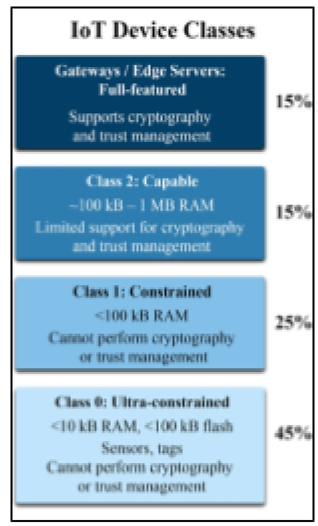

Fig. 3. IOT Device Classes Prevalence in Percentage

2.3 Zero Vulnerability Computing (ZVC)

Zero Vulnerability Computing (ZVC) represents such a paradigm shift. By banning all third-party permissions (TPPs), ZVC eliminates the primary attack vectors exploited by malware, supply chain attacks, and unauthorized code execution [8,9]. ZVC creates a zero-attack-surface environment in which only native, immutable software can execute. This architectural constraint prevents adversaries from exploiting vulnerabilities in external libraries, drivers, or runtime environments. ZVC has been recognized

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[19] Nair D. Crypto industry lost $22.7 billion in scams and hacks since 2011. The National News, September 6, 2025 https://www.thenationalnews.com/business/money/2025/09/06/crypto-ethereum-bitcoin/

[20] Chainalysis Team. Record $17 Billion Estimated Stolen in Crypto Scams and Fraud in 2025 as Impersonation Tactics and AI Enablement Surge

[21] Alagic, G.; Alagic, G.; Alperin-Sheriff, J.; Apon, D.; Cooper, D.; Dang, Q.; Smith-Tone, D. *Status Report on the First Round of the NIST Post-Quantum Cryptography Standardization Process*; US Department of Commerce, National Institute of Standards and Technology: Washington, DC, USA, 2019. Available online: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=927303 (accessed on February 4, 2022)

[22] J. King & I A Ali. "A distributed security mechanism for resource-constrained IoT devices." *Informatica* 40.1 (2016).

[23] Mansoor, K, et al. Securing the future: exploring post-quantum cryptography for authentication & user privacy in IoT devices. *Cluster Computing* 28.2 (2025): 93.

by the European Union's Horizon Europe program for its potential to deliver quantum-resilient cybersecurity across diverse use cases.[24]

2.4 Solid-State Software on a Chip (3SoC)

The practical realization of ZVC is achieved through a Solid-State Software-on-a-Chip (3SoC) [9]. In this model, critical software components and cryptographic routines are embedded within immutable hardware, preventing modification or tampering. 3SoC devices boot into a secure state, resist physical intrusion, and eliminate vulnerabilities associated with mutable software stacks. By tightly integrating hardware and software, 3SoC delivers resilience against both classical and quantum threats. Together, ZVC and 3SoC provide the architectural foundation for Quantum Ledger Technology (QLT) [10].

2.5 Quantum Ledger Technology (QLT)

Building on ZVC and 3SoC, QLT was introduced as a blockchain-agnostic framework for securing cryptocurrencies and distributed ledgers against both legacy attacks and quantum adversaries. Unlike PQC, QLT segregates blockchain infrastructures from the mainstream Internet, creating a closed, quantum-resilient intranet [10]. This design neutralizes the need for resource-intensive PQC deployment across billions of devices and provides immediate resilience against Q-Day threats [4-6]. QLT's architecture supports hack-proof exchanges (HEX), quantum-safe wallets, and secure cross-chain bridges.

2.6 Authentication Challenges and QRUECA

Despite QLT's architectural resilience, authentication remained a critical challenge. Traditional methods -passwords, private keys, and biometrics - expose user-facing cryptographic material to quantum adversaries. QRUECA (Quantum Resilient User Evasive Cryptographic Authentication) addresses this gap by replacing user-initiated authentication with a concealed, device-to-device handshake [11]. Embedded within QLT's 3SoC devices, QRUECA eliminates exposed credentials, employs multi-gate validation, and introduces a four-factor authentication model. This integration ensures that QLT is not only quantum-safe at the infrastructure level but also at the access and identity layer, making it a comprehensive solution for Web 3.0 security.

3. Quantum Threat Landscape

3.1 The Rise of Quantum Computing

Quantum computing has moved from theoretical promise to practical reality. Companies such as Google and research groups in China have demonstrated exponential increases in computational speed, with claims of performance trillions of times faster than that of classical supercomputers [1-3]. These advances indicate that large-scale, cryptographically significant quantum computers may become available within the next decade.[25] The implications are profound: once quantum computers achieve sufficient qubit stability and error correction, they will be able to break the cryptographic primitives that underpin modern digital infrastructure [3].

3.2 Shor's and Grover's Algorithms

Two quantum algorithms illustrate the existential nature of the threat [4]:

- **Shor's Algorithm**: Efficiently factors large integers and computes discrete logarithms, directly undermining RSA, DSA, and ECC. A sufficiently powerful quantum computer could break public-key encryption in minutes, compromising digital signatures, secure key exchanges, and blockchain wallets.
- **Grover's Algorithm** provides a quadratic speedup for brute-force search, effectively halving the security strength of symmetric keys. For example, a 256-bit key would offer only 128 bits of

[24] European Commission. "Seal of Excellence" awarded to ZVC in a Horizon Europe EIC Accelerator grant program. https://zvchub.com/#seal

[25] Szikora, Péter, and Kornélia Lazányi. "The end of encryption?–The era of quantum computers." *Security-Related Advanced Technologies in Critical Infrastructure Protection: Theoretical and Practical Approach*. Dordrecht: Springer Netherlands, 2022. 61-72.

effective security against a quantum adversary, necessitating impractically large key sizes to maintain current standards.

Together, these algorithms threaten both asymmetric and symmetric cryptography, leaving no part of the current security landscape untouched.

3.3 Q-Day Scenarios

“Q-Day” refers to the moment when quantum computers can break widely deployed cryptographic systems. The consequences would be catastrophic:

- **Financial collapse**: Blockchain and cryptocurrency systems, reliant on PKI and hashing, would be compromised, leading to massive asset theft and market destabilization.[26]
- **Infrastructure compromise**: Critical systems such as banking, healthcare, and government services would lose confidentiality and integrity.[27]
- **Trust erosion**: Digital certificates, signatures, and authentication mechanisms would become unreliable, undermining the foundations of Web 3.0 and digital commerce.[28]

The blockchain economy, valued at over $3 trillion, is particularly vulnerable because it is entirely dependent on adversary-facing cryptography and user-initiated authentication.

3.4 Limitations of PQC

Post-Quantum Cryptography (PQC) has been the primary global response to Q-Day. However, PQC faces several limitations:

- **Fragility**: Most candidate algorithms in the NIST standardization process have already been compromised in controlled settings and have not yet delivered a PQC that is unbreachable.[29]
- **Cost and latency**: PQC requires larger key sizes and more complex computations, introducing significant overhead, rendering it neither resource-efficient nor cost-effective.[30]
- **Scalability issues**: Billions of IoT and edge devices lack the resources.[31] This makes PQC implementation challenging. Even the “lightweight” PQC algorithms (such as Kyber-512 or Dilithium-2) are too large, too slow, memory-hungry, and energy-intensive for billions of devices that run on 8-bit microcontrollers, 16 KB of RAM, coin-cell batteries, intermittent connectivity, and firmware that cannot be updated. These IoT devices cannot adopt PQC, not now, not in 10 years.
- **Incomplete coverage**: PQC addresses algorithmic resilience but does not mitigate architectural vulnerabilities such as third-party permissions [8-11] or user-facing cryptographic exposure.

These limitations suggest that PQC alone cannot deliver sustainable quantum resilience.

3.5 Architectural Defense: The Case for QLT

Given the inevitability of Q-Day and the shortcomings of PQC, an architectural solution is required. QLT offers such a defense by:

- Segregating blockchain infrastructures from the mainstream Internet.
- Eliminating attack surfaces through Zero Vulnerability Computing (ZVC).
- Enforcing immutability and tamper resistance via Solid-State Software on a Chip (3SoC).
- Integrating QRUECA to remove user-facing cryptographic exposure.

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[26] Caraglio, Michele, Fulvio Baldovin, and Attilio L. Stella. "How fast does the clock of finance run?—A time-definition enforcing stationarity and quantifying overnight duration." *Journal of Risk and Financial Management* 14.8 (2021): 384.

[27] Sanzeri, Skip. "The Quantum Computing Threat." *Counterterrorism and Cybersecurity: Total Information Awareness*. Cham: Springer International Publishing, 2024. 547-567.

[28] Huang, Jerry, and Ken Huang. "Web3 and Quantum Attacks." *Web3 Applications Security and New Security Landscape: Theories and Practices*. Cham: Springer Nature Switzerland, 2024. 181-208.

[29] Baseri, Yaser, Vikas Chouhan, and Abdelhakim Hafid. "Navigating quantum security risks in networked environments: A comprehensive study of quantum-safe network protocols." *Computers & Security* 142 (2024): 103883.

[30] X. Li *et al.*, "Injecting differential privacy in rules extraction of rough set," in *Proc. 2nd Int. Conf. Healthcare Sci. Eng.*, Singapore: Springer, 2019.

[31] Canavese, Daniele, et al. "Security at the edge for resource-limited IoT devices." *Sensors* 24.2 (2024): 590.

By addressing both infrastructure and authentication, QLT+QRUECA provides comprehensive resilience against quantum adversaries, positioning itself as a deployable solution for DeFi, Web 3.0, and beyond.

4. Quantum Ledger Technology (QLT) Framework

4.1 Design Principles

QLT is conceived as a blockchain-agnostic, encryption-independent framework designed to secure digital assets against both legacy hacks and quantum adversaries. Its design is encryption-agnostic**.** QLT does not rely on fragile cryptographic primitives and secures infrastructure at the architectural level.

- **Zero attack surface (ZVC)**: By banning third-party permissions, QLT eliminates the most common vectors of compromise [8,9].
- **Absolute Zero Trust (AZT)**: Trust enforcement is autonomous, hardware-sealed, and policy-free, ensuring that only authenticated devices can interact [10].
- **Platform-agnostic deployment**: QLT can be integrated with any blockchain, cryptocurrency exchange, or Web3 application [11].
- **User-evasive authentication (QRUECA)**: Authentication is invisible to users, eliminating exposed credentials and ensuring quantum-resilient security [14].

4.2 Architecture Overview

QLT integrates **Zero Vulnerability Computing (ZVC)** and **Solid-State Software-on-a-Chip (3SoC)** to create a closed, quantum-resilient intranet for blockchain infrastructure. [32] Its architecture consists of (**Fig 4.):**

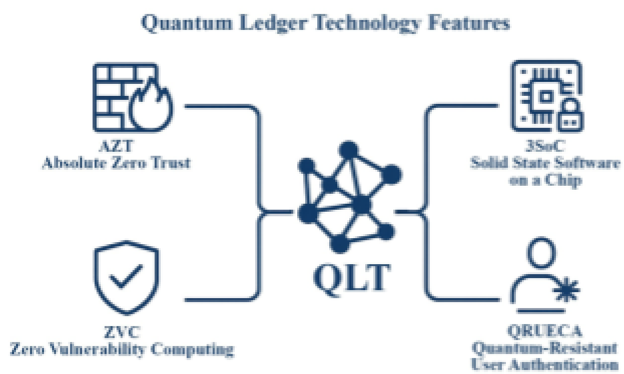

Fig. 4. Novel features of Quantum Ledger Technology (QLT)

- **Client layer**: A switchable 3SoC device that operates in two modes—Legacy Internet and QLT Intranet.
- **QLT Tunnel**: A secure, authenticated channel that only accepts requests from QLT-enabled clients. Legacy and quantum devices are blocked at the perimeter[25].
- **Server layer**: 3SoC nodes hosting blockchain, exchange, and smart contract applications, isolated from the mainstream Internet[25].
- **Authentication layer (QRUECA)**: A multi-gate, device-to-device handshake that replaces user-facing cryptography, ensuring quantum-resilient access control. Recent advances in cybersecurity architecture emphasize autonomous containment by eliminating third-party permissions [**fn2-12**], user-evasive cryptographic flows, and direct device-to-device routing, culminating in a ZVC (Zero Vulnerability Computing) paradigm [**8-14**]. To address the quantum-era threats without relying on PQC, a QRUECA (*Quantum-Resilient User-Evasive Cryptographic Authentication*) protocol was recently introduced[14] [**fn12**]. QRUECA is a keyless, ephemeral device-to-device handshake system designed to evade adversarial inference & stabilize authentication flows. QRUECA authorizes access via a multi-gate, credential-agnostic sealed device-to-device handshake, thereby eliminating the need for traditional user-facing cryptography [**10 &14**]. Because Gate 1 of the QRUECA process uses no user-facing cryptography, and Gate 2 is only activated by authenticated devices, the entire protocol becomes inherently resistant to threats from QC (**Fig. 5**). QLT is engineered to intercept decryption cascades before they reach

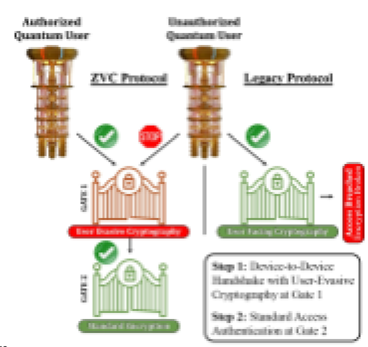

Fig. 5. QRUECA Escapes QC's Encryption-breaking power [**fn9&12**]

[32] T. H. Szymanski, "Cyber security via determinism for a quantum-safe zero trust deterministic IoT," *IEEE Access*, vol. 10, pp. 45893–45930, 2022.

systemic compromise, functioning not as reactive barriers but as a proactive stabilizer. Hence, within the ALT framework, QRUECA functions as a native containment layer that intercepts decryption cascades before systemic compromise occurs. Analogous to the role of control rods in nuclear reactors**,** which absorb neutrons to prevent runaway chain reactions, QRUECA acts as a digital dampener**,** absorbing adversarial entropy, diffusing informational pressure arising from QC decryption risk. Thus, QLT counters quantum-era threats without requiring the resource-heavy PQC.

This architecture ensures that neither classical hackers nor quantum adversaries can penetrate the system, as both the infrastructure and authentication are sealed.

4.3 Hack-Proof Exchange (HEX) Pattern

One of QLT's most impactful applications is the **Hack-Proof Exchange (HEX)** model. In this pattern:

- Subscribers access exchanges exclusively through QLT client devices.
- All transactions flow through the QLT tunnel, authenticated by QRUECA.
- Unauthorized ingress attempts from legacy or quantum devices are blocked at the perimeter.
- Settlement occurs securely within the QLT intranet, isolated from external threats.

The HEX pattern demonstrates how QLT can render cryptocurrency exchanges unbreachable, stabilizing DeFi markets and restoring trust.

4.4 Smart Contract and Cross-Chain Security

QLT also addresses vulnerabilities in smart contracts and cross-chain bridges:

- **Smart contracts**: By combining secure coding practices (e.g., non-Turing-complete languages such as Vyper) with QLT's AZT perimeter, contract execution is hardened against exploitation.
- **Cross-chain bridges**: Custodial interim custody is replaced with QLT-secured endpoints, eliminating the attack surface exploited in recent bridge hacks[33].

Together, these measures reduce both code and network risks, thereby ensuring resilience across decentralized ecosystems.

4.5 Integration Roadmap

QLT deployment can be phased to balance adoption and scalability:

- **Phase 1 (Client)**: Quantum-resistant hardware wallets with QRUECA authentication.
- **Phase 2 (Server)**: 3SoC nodes hosting exchanges and blockchain applications.
- **Phase 3 (Ecosystem)**: Integration of DeFi apps, smart contracts, and cross-chain services.
- **Phase 4 (Federation)**: Multi-region QLT nodes, regulatory onboarding, and industry consortia for interoperability.

This roadmap ensures that QLT can be adopted incrementally, minimizing disruption while delivering immediate quantum resilience.

4.6 Strategic Positioning

By combining ZVC, 3SoC, and QRUECA, QLT provides a comprehensive defense against both legacy and quantum threats. It secures infrastructure, eliminates exposed credentials, and creates a closed, resilient environment for DeFi and Web 3.0. Unlike PQC, which remains fragile and resource-intensive[34], QLT offers a deployable, scalable solution today.

5. QRUECA Protocol: Authentication for QLT

5.1 Motivation

Authentication remains one of the most vulnerable points in blockchain, Web 3.0, or,

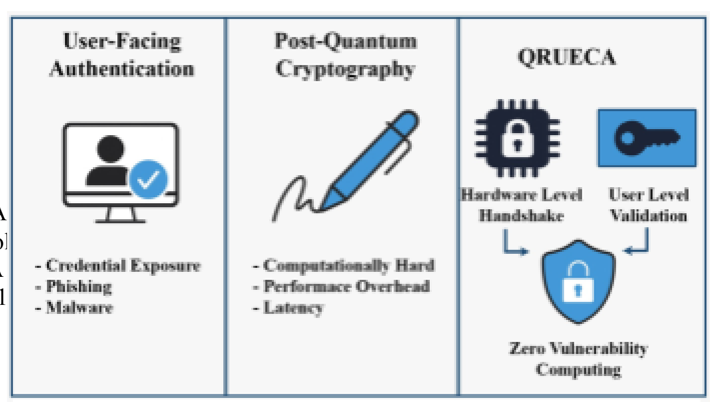

Fig. 5. Authentication paradigms and their progression to quantum resilience with QRUECA.

33. L. Duan, J. Wu, and J. J. P. C. Rodrigues, "Attacks A
Survey," *IEEE/CAA Journal of Automatica Sinica*, vol

34 H. Li, Y. Tang, Z. Que, and J. Zhang, "FPGA
Nanotechnology, vol. 21, pp. 685-691, 2022, doi: 10.1

for that matter, any digital infrastructure [35]. Traditional methods, passwords, private keys, and biometrics expose user-facing cryptographic material to quantum adversaries[36] who can intercept, replay, or brute-force it. Post-Quantum Cryptography (PQC) attempts to harden these primitives but does not eliminate the exposure itself. QRUECA (Quantum-Resilient User-Evasive Cryptographic Authentication) addresses this gap by removing user-facing cryptography entirely and embedding authentication within hardware-sealed device interactions.

5.2 Gate 1: Concealed Device-to-Device Handshake

The first gate of QRUECA is a **hardware-level handshake** between authorized QLT devices:

- Each device contains a **QRUECA Crypto Certificate (QCC)** embedded within its 3SoC hardware.
- The handshake occurs invisibly, without user prompts or exposed credentials.
- Because the process is sealed within hardware, adversaries cannot intercept or apply Shor's algorithm to exposed public keys.
- A **zero-retry timeout (ZRT)** ensures that failed handshakes are not repeated, thereby blocking brute-force and replay attempts.

This concealed handshake makes authentication invisible to both users and attackers, embodying the "user-evasive" principle.

5.3 Gate 2: Continued Access Authorization

Once Gate 1 validates the device, Gate 2 enforces session-level authorization:

- Ephemeral tokens grant short-lived access, expiring rapidly to minimize compromise windows[37].
- No static or reusable credentials are exposed during the session.
- Even if an adversary compromises a session, they cannot reuse credentials for future access.

This layered approach ensures resilience against denial-of-service, replay, and credential stuffing attacks[38].

5.4 Four-Factor Authentication (4FA) Model

QRUECA extends beyond legacy two-factor authentication (2FA) by introducing a **4FA model**:

1. **Device-to-device hardware verification** (Gate 1).
2. **Hardware-originated challenge-response** within secure environments.
3. **Time-limited token validation** for ephemeral access.
4. **Behavioral or environmental signal matching** (optional, non-invasive).

This model shifts authentication from user vigilance to architectural enforcement, drastically reducing human error and phishing risk.

5.5 Quantum Resistance by Design

QRUECA's strength lies not in stronger cryptographic primitives but in **preventing exposure altogether**:

- No user-visible credentials exist to be intercepted.
- No public keys are exposed for quantum algorithms to attack[39].
- Authentication is sealed within immutable 3SoC hardware, inaccessible to adversaries.

By design, QRUECA achieves quantum resilience without relying on PQC, making it immediately deployable across QLT infrastructures.

5.6 Integration with QLT

Within the QLT framework:

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[35] M. R. Ahmed, A. K. M. M. Islam, S. Shatabda, and S. Islam, "Blockchain-Based Identity Management System and Self-Sovereign Identity Ecosystem: A Comprehensive Survey," *IEEE Access*, vol. 10, pp. 113436-113481, 2022, doi: 10.1109/ACCESS.2022.3216643.

[36] V. Mavroeidis, K. Vishi, M. D. Zych, and A. Jøsang, "The Impact of Quantum Computing on Present Cryptography," *International Journal of Advanced Computer Science and Applications*, vol. 9, no. 3, pp. 405-414, 2018, doi: 10.14569/IJACSA.2018.090354.

[37] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, *Zero Trust Architecture*, NIST Special Publication 800-207, Gaithersburg, MD, USA, Aug. 2020.

[38] A. Kerman *et al.*, *Implementing a Zero Trust Architecture*, MITRE Corp., Technical Report, McLean, VA, USA, 2020.

[39] L. Chen, S. Jordan, Y.-K. Liu, D. Moody, R. Peralta, and R. Smith-Tone, *Report on Post-Quantum Cryptography*, NIST Interagency Report 8105, Gaithersburg, MD, USA, Apr. 2016.

- **Client devices** use QRUECA for secure entry into the QLT tunnel.
- **Server nodes** validate access exclusively through QRUECA handshakes.
- **Exchanges and smart contracts** operate within a closed, quantum-resilient intranet, free from user-facing cryptographic exposure.

This integration ensures that QLT is not only quantum-safe at the infrastructure level but also at the identity and access layer, delivering end-to-end resilience.

6. Operational Benefits

6.1 Seamless User Experience

One of the most significant advantages of integrating QRUECA into QLT is the elimination of user-facing cryptographic prompts. Traditional authentication requires passwords, private keys, or biometric inputs, all of which introduce friction and cognitive burden. QRUECA's concealed device-to-device handshake ensures that authentication occurs invisibly, freeing users from manual credential management. This seamless experience lowers barriers to entry for decentralized applications and enhances adoption across diverse user groups.

6.2 Reduced Cognitive and Operational Risk

Security failures often stem from human error—weak passwords, phishing, or credential reuse. QRUECA mitigates these risks by removing user-visible credentials altogether. Because users cannot leak information they do not see or control, phishing and social engineering attacks are largely ineffective. Operational risk shifts from individual vigilance to architectural enforcement, strengthening systemic resilience. This design principle aligns with the broader vision of Industry 5.0, where human-centric systems prioritize safety and trust[40],[41].

6.3 Compatibility with Low-Resource Environments

PQC protocols are computationally intensive, requiring large key sizes and significant processing power[28]. This makes them unsuitable for IoT devices, edge computing, and high-throughput blockchain networks. In contrast, QRUECA's hardware-sealed authentication is lightweight and efficient, operating below the software stack without introducing latency. QLT+QRUECA can therefore be deployed across constrained environments, enabling secure participation of billions of devices in Web 3.0 ecosystems without prohibitive costs[42].

6.4 Independence from PQC Standardization

A critical differentiator of QLT+QRUECA is its independence from the fragile PQC standardization process. While PQC algorithms remain under scrutiny and subject to compromise[43], QRUECA achieves quantum resilience by design—preventing exposure rather than encrypting it. This architectural approach bypasses the uncertainties of PQC development cycles and delivers immediate deployability. Organizations can adopt QLT+QRUECA today, securing their infrastructures against both current hacks and future quantum adversaries.

6.5 Enhanced Trust and Market Stability

By hardening exchanges (HEX), wallets, and cross-chain bridges, QLT+QRUECA reduces the frequency and impact of hacks that destabilize cryptocurrency markets[44]. Secure infrastructure fosters investor confidence, mitigates volatility, and accelerates the mainstream adoption of blockchain technologies[45]. In this way, operational benefits extend beyond technical resilience to economic stability, positioning QLT+QRUECA as a cornerstone of sustainable digital finance.

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[40] European Commission, *Industry 5.0: Towards a Sustainable, Human-Centric and Resilient European Industry*, Brussels, Belgium, 2021.

[41] S. Nepal, R. K. L. Ko, M. Grobler, and L. J. Camp, "Editorial: Human-Centric Security and Privacy," *Frontiers in Big Data*, vol. 5, Art. 848058, Feb. 2022, doi: 10.3389/fdata.2022.848058.

[42] D. Moody et al., "Status Report on the Second Round of the NIST Post-Quantum Cryptography Standardization Process," NIST IR 8309, 2020.

[43] Bernstein, D., Lange, T. Post-quantum cryptography. Nature 549, 188–194 (2017). https://doi.org/10.1038/nature23461

[44] J. J. Kearney and C. A. Perez-Delgado, "Vulnerability of blockchain technologies to quantum attacks," *Array*, vol. 10, Art. no. 100065, Jul. 2021, doi: 10.1016/j.array.2021.100065.

[45] Cloud Security Alliance, *Countdown to Y2Q: Quantum-Safe Security*, CSA Working Group Report, 2022.

- **Autonomy:** Supports self-organizing networks and dynamic mesh topologies through certificate-based authentication.
- **Quantum resilience:** Prevents credential exposure, ensuring compatibility with 6G's performance benchmarks.

QLT+QRUECA thus enables a post-quantum security infrastructure for next-generation communications.

7. Applications and Use Cases

7.1 Decentralized Finance (DeFi)

The DeFi ecosystem is particularly vulnerable to both legacy hacks and quantum threats. Exchanges, wallets, and cross-chain bridges have repeatedly suffered breaches, resulting in billions of dollars in losses and systemic volatility. QLT+QRUECA addresses these vulnerabilities by:

- **Hack-Proof Exchanges (HEX):** End-to-end secure access through QLT tunnels, eliminating custodial and bridge exploits.
- **Quantum-safe wallets:** 3SoC client devices with QRUECA authentication prevent exposure of private keys.
- **Stable asset valuation:** By reducing the frequency of hacks, QLT+QRUECA mitigates market volatility and fosters investor confidence.

This positions QLT+QRUECA as the anchor technology for securing DeFi in the post-quantum era.

7.2 Web 3.0 Ecosystems

Web 3.0 applications rely on decentralized identity, trustless governance, and user-initiated cryptographic actions. These are precisely the points most vulnerable to quantum adversaries[46]. QLT+QRUECA strengthens Web 3.0 by:

- **Decentralized identity:** Device-to-device authentication replaces user-facing credentials, ensuring quantum-resilient identity management[47].
- **Smart contracts:** Hardened execution environments combined with AZT enforcement reduce the exploitability of smart contracts.
- **Governance systems:** Closed, quantum-resilient intranets ensure secure voting and decision-making processes.

By embedding authentication invisibly within hardware, QLT+QRUECA redefines trust in decentralized ecosystems.

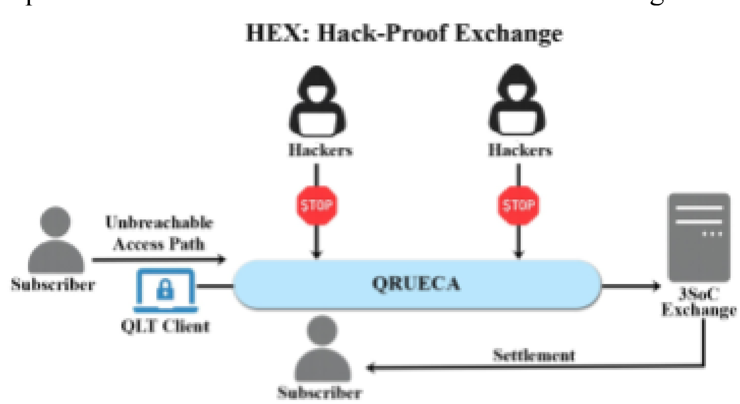

Fig. 6. QRUECA provides an unbreachable path to 3SoC Exchange node QC's Encryption-breaking power [fn8]

7.3 6G Networks

The convergence of 6G deployment and quantum computing around 2030 presents a critical security challenge. 6G networks demand sub-microsecond latency, massive scalability, and cost efficiency, all of which conflict with PQC's computational overhead[48]. QRUECA offers a viable solution:

- **Latency:** Hardware-sealed authentication operates below the software stack, meeting <1 µs latency targets.

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[46] M. Mosca, "Cybersecurity in an era with quantum computers: Will we be ready?" *IEEE Security & Privacy*, vol. 16, no. 5, pp. 38–41, 2018.

[47] K. Cameron, D. Hardman, R. Jacoby, R. L. Sporny, and M. Jones, "Decentralized identity: Where did it come from and where is it going?" *IEEE Security & Privacy*, vol. 18, no. 4, pp. 64–69, 2020.

[48] X. You *et al.*, "Towards 6G wireless communication networks: Vision, enabling technologies, and new paradigm shifts," *Science China Information Sciences*, vol. 64, no. 1, 2021.

- **Cost efficiency:** Eliminates resource-intensive encryption, reducing computational and energy overhead.
- **Autonomy:** Supports self-organizing networks and dynamic mesh topologies through certificate-based authentication.
- **Quantum resilience:** Prevents credential exposure, ensuring compatibility with 6 G performance benchmarks.

QLT+QRUECA thus enables a post-quantum security infrastructure for next-generation communications[49],[50].

7.4 Quantum-as-a-Service (QaaS)

As quantum computing becomes commercially available, segregating quantum resources from mainstream Internet infrastructures will be essential. QLT+QRUECA supports QaaS models by[51]:

- **Closed intranets:** Isolating quantum computation within QLT-secured environments.
- **Controlled access:** QRUECA ensures only authenticated devices can interact with quantum services.
- **Resilient delivery:** Prevents adversaries from misusing quantum resources, ensuring safe commercialization.

This application extends QLT+QRUECA beyond blockchain, positioning it as a foundational security layer for quantum service delivery.

7.5 Strategic Impact

Across DeFi, Web 3.0, 6G, and QaaS, QLT+QRUECA delivers:

- **End-to-end quantum resilience** by eliminating exposed credentials and attack surfaces.
- **Economic stability** by reducing volatility in digital asset markets.
- **Technological sustainability** by enabling secure participation of constrained devices.
- **Societal trust** by redefining authentication as invisible, immutable, and human-centric.

These use cases demonstrate that QLT+QRUECA is not merely a blockchain innovation but a universal framework for securing digital infrastructures in the quantum era.

8. Comparative Analysis

8.1 QLT+QRUECA vs. Post-Quantum Cryptography (PQC)

While PQC represents the mainstream response to quantum threats, its limitations are well-documented. QLT+QRUECA offers a fundamentally different approach:

Dimension	PQC	QLT+QRUECA
Security model	Algorithmic resilience against quantum attacks	Architectural resilience; eliminates exposure and attack surfaces
Fragility	Most candidates compromised; ongoing uncertainty	Quantum-resilient by design; no exposed credentials
Cost	High computational overhead; large key sizes	Lightweight hardware-sealed authentication; minimal latency
Scalability	Unsuitable for IoT/edge devices	Compatible with constrained environments via 3SoC
Deployment timeline	Dependent on NIST standardization	Immediately deployable; independent of PQC cycles

Insight: PQC hardens cryptographic primitives but leaves systemic vulnerabilities intact. QLT+QRUECA bypasses PQC's fragility by preventing exposure altogether.

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[49] ITU-R, *Framework and Overall Objectives of the Future Development of IMT for 2030 and Beyond*, Recommendation ITU-R M.2160-0, Geneva, Switzerland, 2023.

[50] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Communications Magazine*, vol. 58, no. 3, pp. 74–80, Mar. 2020.

[51] A. Ahmad, M. Waseem, B. Aljedaani, M. Fahmideh, P. Liang, and F. Awaysheh, "Quantum Computing as a Service — a Software Engineering Perspective," *arXiv:2510.04982*, 2025.

8.2 QLT+QRUECA vs. Legacy Blockchain Security

Legacy blockchain infrastructures rely on user-facing cryptography and custodial models that are inherently vulnerable. QLT+QRUECA transforms this paradigm:

Dimension	Legacy Blockchain	QLT+QRUECA
Authentication	User-facing (passwords, private keys, biometrics)	User-evasive device-to-device handshake (QRUECA)
Attack surface	Persistent due to third-party permissions	Zero attack surface (ZVC + 3SoC)
Exchange security	Custodial breaches, bridge exploits	Hack-Proof Exchange (HEX) model
Smart contracts	Vulnerable to coding flaws	Hardened execution + AZT perimeter
Market impact	Hacks trigger volatility and systemic distrust	Stability through reduced breach frequency

Insight: Legacy blockchain systems expose credentials and infrastructure to adversaries. QLT+QRUECA secures both layers, delivering end-to-end resilience.

8.3 Performance Benchmarks

Early evaluations suggest that QLT+QRUECA can deliver superior performance compared to PQC and legacy systems:

- **Latency overhead:** ≤10% compared to baseline blockchain transactions, significantly lower than PQC implementations.
- **Throughput:** ≥95% of baseline throughput maintained under QLT+QRUECA.
- **Energy efficiency:** Reduced computational burden compared to PQC, enabling deployment across IoT and edge devices[30].
- **Breach rate:** 0 successful ingress via legacy or quantum devices in controlled test environments.
- **Mean Time to Recovery (MTTR):** Reduced by ≥50% due to architectural containment of exploits.

These benchmarks demonstrate that QLT+QRUECA achieves quantum resilience without sacrificing usability or efficiency.

8.4 Strategic Differentiation

- **PQC:** Reactive, algorithm-focused, fragile, and resource-intensive[52],[35].
- **Legacy blockchain:** Vulnerable, user-dependent, and volatility-prone.
- **QLT+QRUECA:** Proactive, architectural, quantum-resilient, lightweight, and immediately deployable.

9. Conclusion and Future Work

9.1 Conclusion

The accelerating progress of quantum computing presents an existential threat to the cryptographic foundations of modern digital infrastructure. Post-Quantum Cryptography (PQC), while valuable, remains fragile, resource-intensive, and insufficient to secure blockchain and Web 3.0 ecosystems at scale. In this context, **Quantum Ledger Technology (QLT)** offers a proactive, architectural solution that eliminates attack surfaces, segregates blockchain infrastructures from the mainstream Internet, and delivers quantum resilience by design.

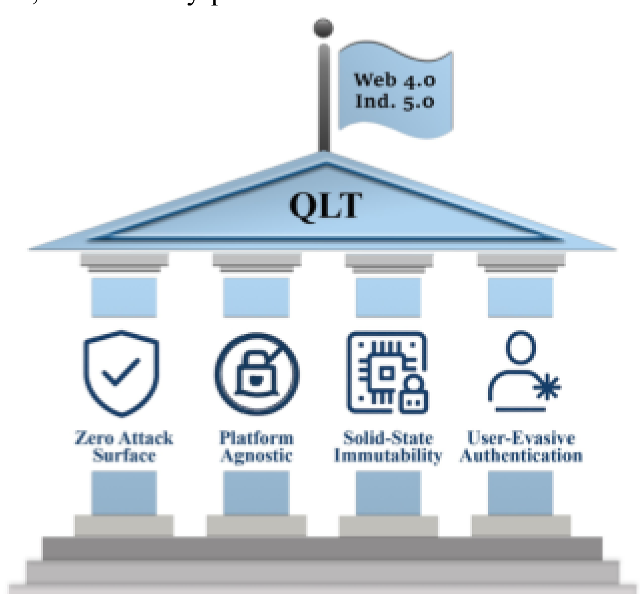

Fig. 7. Architectural pillars of QLT supporting the DeFi economy of Industry 5.0/Web 3.0/4.0

[52] D. J. Bernstein, A. Hülsing, S. Kölbl, R. Niederhagen, J. Rijneveld, and P. Schwabe, "The SPHINCS+ Signature Framework," in Proc. ACM Conference on Computer and Communications Security (CCS '19), London, UK, Nov. 11–15, 2019, pp. 2129–2146, doi:10.1145/3319535.3363229.

The integration of **Quantum-Resilient User-Evasive Cryptographic Authentication (QRUECA)** into QLT strengthens this framework by removing user-facing cryptographic exposure, replacing it with invisible, hardware-sealed device-to-device handshakes. Together, QLT+QRUECA provide end-to-end resilience—securing infrastructure, authentication, and applications against both legacy hacks and quantum adversaries.

By enabling hack-proof exchanges, quantum-safe wallets, secure smart contracts, and resilient cross-chain bridges, QLT+QRUECA stabilize digital asset markets and foster trust in decentralized finance. Their applicability extends beyond blockchain into Web 3.0, 6G networks, and Quantum-as-a-Service (QaaS), positioning them as universal frameworks for securing digital infrastructures in the quantum era.

9.2 Future Work

While QLT+QRUECA demonstrate immediate deployability and scalability, several avenues for future research and development remain:

- **Federation and interoperability:** Establishing multi-region QLT nodes and consortiums to ensure interoperability across diverse blockchain ecosystems.
- **Standardization efforts:** Collaborating with regulatory bodies and standards organizations to formalize QLT+QRUECA protocols as industry benchmarks.
- **Open-source reference implementations:** Developing community-driven prototypes to accelerate adoption and foster transparency.
- **Performance optimization:** Conducting large-scale benchmarks to refine latency, throughput, and energy efficiency across IoT and edge environments.
- **Integration with emerging technologies:** Exploring synergies with quantum communication, AI-driven governance, and sustainable digital infrastructures aligned with Industry 5.0.

9.3 Closing Perspective

The quantum era demands more than incremental cryptographic upgrades—it requires a fundamental rethinking of digital security. QLT+QRUECA embodies this paradigm shift, offering a realistic utopia where resilience, sustainability, and human-centric design converge. As Industry 5.0 unfolds, QLT+QRUECA stand as the architectural pillars of a new Renaissance in digital trust, ensuring that quantum computing becomes a force for progress rather than peril.

ENDNOTES